A

Interface

Individual-Level Data

`susie(X,y)`

Summary Statistics

Ordinary RSS

`susie_rss(z,R)`

`susie_rss(z,X)` ★

Stochastic LD

`susie_rss(z,R,B>0)` ★

`susie_rss(z,X,B=TRUE)` ★

Regularized LD ($\lambda>0$)

`susie_rss(z,R,λ>0)`

`susie_rss(z,X,λ>0)` ★

Mixture of LD References

`susie_rss(z, X=list(X_1,…,X_K), B=TRUE,λ>0)` ★

Constructor

Data Constructors ★

Validate Inputs & Build S3-Typed Data Objects

`individual_constructor()`

`ss_constructor()`

`rss_constructor()`

S3 Backend

Method Dispatch ★

Resolved by S3 Data Class

`<method>.individual()`

`<method>.ss()`

`<method>.rss_lambda()`

mvsusieR

Adopts susieR 2.0 Backend

`<method>.mv_ss()`

Workhorse

Model Initialization ★

`ibss_initialize(data,params)`

IBSS ★

`ibss_fit(data,params,model)`

mvsusieR *calls via S3 dispatch*

Unmappable Effect Models

SuSiE-ash ★ **SuSiE-inf** ★

SER and Residual Variances

Maximum ELBO

Method of Moments ★

Inverse Gamma Prior ★

Stochastic LD Correction ★

Multi-Panel LD Weight Update

Mixture Weight Optimization ★

Model Finalization ★

`ibss_finalize(data,model)`

Refinement

Block Coordinate Ascent

IBSS Fit Refinement

mvsusieR

Residual variance estimation with missing data in Y

Output

Posterior Inclusion Probabilities

Credible Sets

Effect Size Estimates

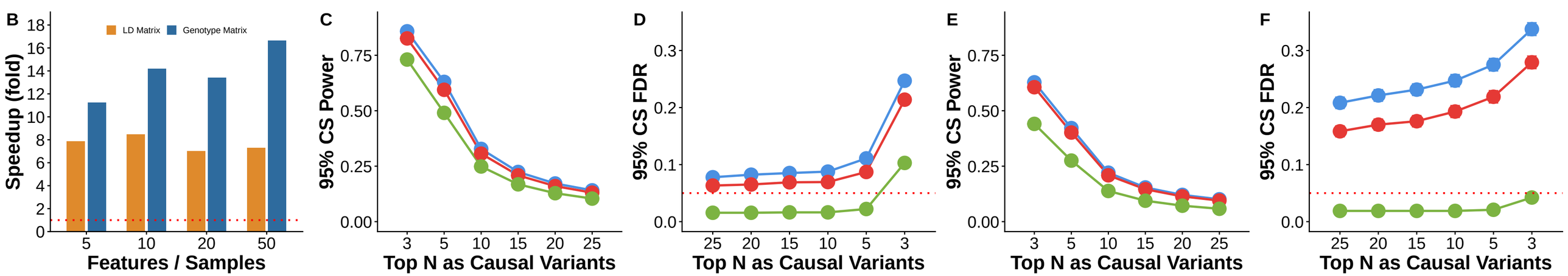
Predictions

Convergence Diagnosis Metrics

★ **New in susieR 2.0**

Enhanced in susieR 2.0

SuSiE

SuSiE-ash

SuSiE-inf